BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: DIK-SSC-X-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 12 | **Paper Code:** DIK-SSC-X-2022 (A)

Q1. Choose the correct option and fill in empty box, given against each statement:

Q1. Which of the following decolourises Br_2 water?

(A) Ethane
(B) Ethene
(C) Propane
(D) Methane

Q2. Carbon is bonded on both sides of oxygen atom in _____

(A) Alcohol
(B) Ketone
(C) Aldehyde
(D) Ether

Q3. Deficiency of vitamins A causes _____

(A) Scurvy
(B) Rickets
(C) Night blindness
(D) Aging

Q4. Permanent hardness of water is removed by adding:

(A) Washing Soda
(B) Soda lime
(C) Caustic soda
(D) Baking soda

Q5. Ozone is present in _____

(A) Mesophese
(B) Thermosphere
(C) Stratosphere
(D) Troposphere

Q6. The density of water is maximum at _____

(A) $-40C^0$
(B) $40C^0$
(C) $-4C^0$
(D) $4C^0$

Q7. Percentage of nitrogen in Urea is _____.

(A) 14 (B) 28

(C) 60 (D) 46

Q8. The impurity present in mineral is called _____

(A) Metal (B) Ore

(C) Ganque (D) Salt

Q9. The salt formed when the hydrogen ion of an acid is completely replaced by metal ions or a group of atom behave like metal ions is called_____

(A) Acidic salt (B) Basic salt

(C) Neutral salt (D) None of these

Q10. The equilibrium constant (kc) expression for the reaction $2H_2 + O_2 \rightleftharpoons 2H_2O$ is _____

(A) $[2H_2O]/[2H_2][O_2]$ (B) $[H_2O]^2/[H_2]^2[O]^2$

(C) $[H_2O]^2/[2H_2][O]2$ (D) $[2H_2][O_2]/[2H_2O]$

Q11. $AlCl_3$ is an acid according to _____ concept.

(A) Arrhenius (B) Lowery and Bronsted

(C) Lewis (D) All of these

Q12. Natural gas is mainly composed of _____

(A) CH_4 (B) C_2H_8

(C) C_2H_6 (D) C_3H[illegible in source scan]

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 21 | **Paper Code:** DIK-SSC-X-2022 (A)

Q2. Attempt any 8 questions of the following:

(i) Calculate the pH of 0.2M solution of NaOH.

(ii) Compare the macroscopic characteristics of forward and reverse reactions.

(iii) Describe the Bronsted Lowery concept of acids and basis.

(iv) Explain the term homologous series.

(v) Justify alkenes and alkynes as unsaturated hydrocarbons.

(vi) Differentiate between fats and oils.

(vii) What are the adverse effects of global warming?

(viii) Write the physical properties of water.

(ix) How blister copper is purified?

(x) What are carbohydrates? Give examples.

(xi) What are cycloalkanes?

SECTION – C

Maximum Marks: 24

> Note: Attempt any 3 questions: Each carry 8 marks.

Q3.

(a) Derive an expression for equilibrium constant of a general reaction.

(b) Distinguish between strong acids and weak acids. Give examples.

Q4.

(a) Write the general characteristics of organic compounds.

(b) How alkenes are prepared by dehydration of alcohols?

Q5.

(a) Sketch the flow sheet diagram of Solvay process.

(b) How are proteins important to living organisms?

Q6.

(a) Distinguish between temporary hard water and permanent hard water.

(b) Describe ozone formation.